



PowerPoint Presentation by
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MANAGEMENT ACCOUNTING

8th EDITION

BY

HANSEN & MOWEN

14 INVENTORY MANAGEMENT

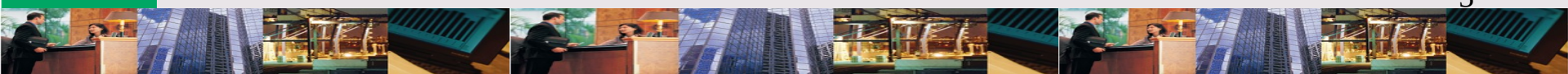
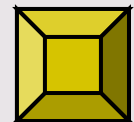
LEARNING OBJECTIVES

After studying this chapter, you should be able to:

LEARNING OBJECTIVES

1. Describe the traditional inventory management model.
2. Discuss JIT inventory management.
3. Explain the theory of constraints (TOC) & tell how it can be used to management inventory.

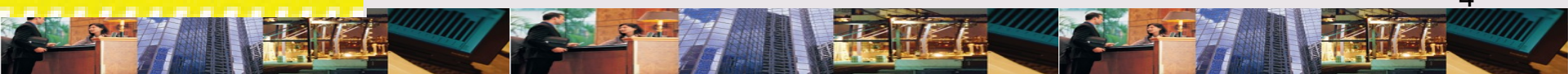
*Click the button to skip
Questions to Think About*



QUESTIONS TO THINK ABOUT: Swasey Trenchers

Why do firms carry inventory?
What are inventory costs?

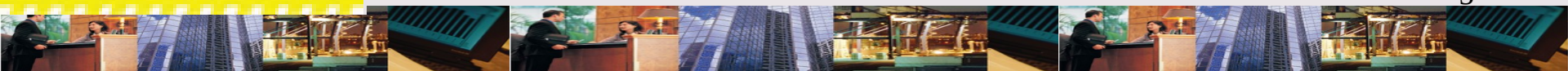
SWASEY



QUESTIONS TO THINK ABOUT: Swasey Trenchers

What can be done to minimize inventory costs? How does JIT reduce inventories?

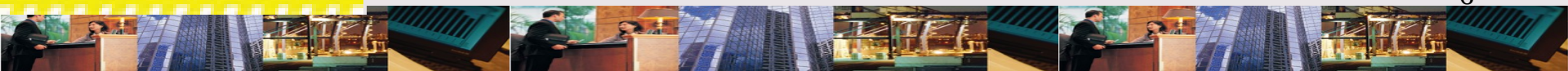
SWASEY



QUESTIONS TO THINK ABOUT: Swasey Trenchers

What are the weaknesses of JIT?
How does using the theory of
constraints reduce inventories?

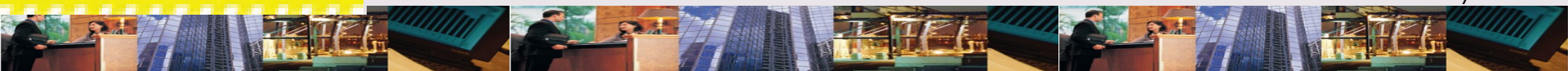
SWASEY



QUESTIONS TO THINK ABOUT: Swasey Trenchers

Why is effective management
of inventory so important?

SWASEY



LEARNING OBJECTIVE

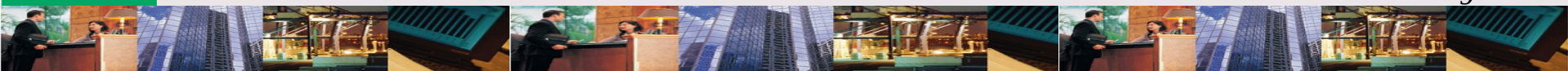
1

Describe the traditional inventory management model.

INVENTORY MANAGEMENT

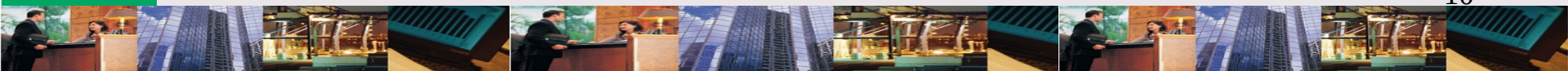
Managing inventory for competitive advantage includes:

- ✧ Quality product engineering
- ✧ Prices
- ✧ Overtime
- ✧ Excess capacity
- ✧ Ability to respond to customers
- ✧ Lead times
- ✧ Overall profitability



INVENTORY COSTS

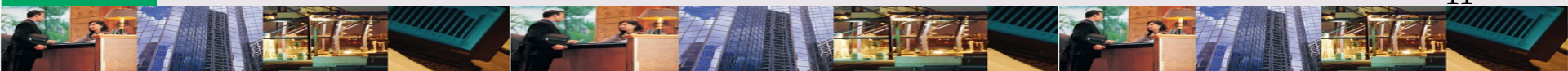
- ✧ Costs to acquire
 - ✧ Ordering costs
 - ✧ Setup costs
- ✧ Carrying costs
- ✧ Stockout costs



HOLDING INVENTORY

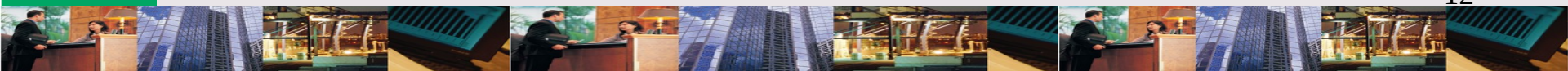
Traditional reasons for holding inventory are:

- ✧ Balancing acquisition & carrying costs
- ✧ Dealing with uncertainty in demand (stockout costs)
- ✧ Creating buffers for needed parts, etc.
- ✧ Producing extra inventory because of unreliable production processes
- ✧ Taking advantage of discounts
- ✧ Hedging against future price increases



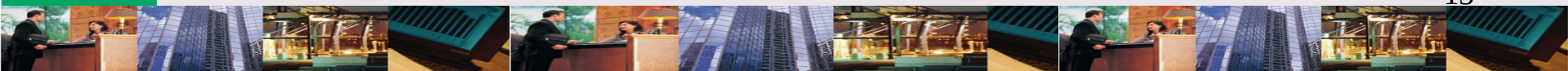
EOQ: Definition

Is a model that calculates the best quantity to order or produce. (**E**conomic **O**rders **Q**uantity)



What are 2 basic questions addressed by EOQ?

- 1. How much should be ordered (produced)?**
- 2. When should the order be placed (setup done)?**



TOTAL COST: Background

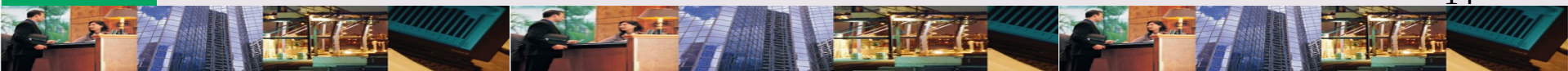
The total cost (TC) formula includes the following:

P = \$25 per order [cost of placing & receiving order (setup & production)]

D = 10,000 [known demand]

Q = 1,000 [order size (or production lot size)]

C = \$2 per unit [carrying cost of 1 unit for 1 year]



FORMULA: Total Cost

Total cost looks at all inventory costs.

Total cost (TC) equation 14.1:

= Ordering cost + Carrying cost

= $PD/Q + CQ/2$

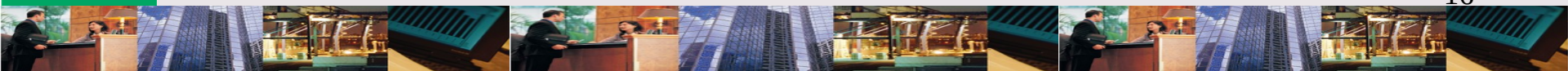
$PD/Q = [(10,000/1,000) \times \$25] = \$ 250$

$CQ/2 = [(1,000/2) \times \$2] = \$1,000$

$TC = \$1,250$

How can the total cost be reduced?

The **EOQ** model will compute the cheapest batch order size.



FORMULA: EOQ

***EOQ** is a calculation intended to lower total inventory costs.*

EOQ equation 14.2:

$$= \sqrt{2 \times \text{Order costs} \div \text{Unit cost}}$$

$$= \sqrt{2PD/C}$$

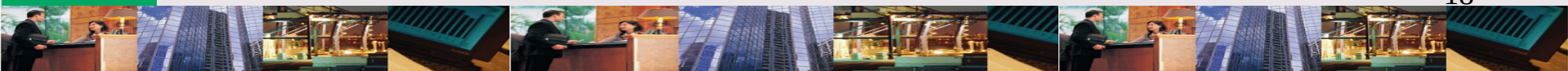
$$= \sqrt{2 \times \$25 \times 10,000 / \$2}$$

$$= \sqrt{250,000}$$

$$= 500$$

What do you do with the order quantity calculated by the **EOQ** model?

Enter the order quantity into the TC equation in **14.1**.



FORMULA: EOQ Cost

EOQ Total cost calculates TC using the EOQ batch size in units to cut total cost by \$250.

Total cost (TC) equation 14.1:

= Ordering cost + Carrying cost

= $PD/Q + CQ/2$

$PD/Q = [(10,000/500) \times \$25] = \$ 500$

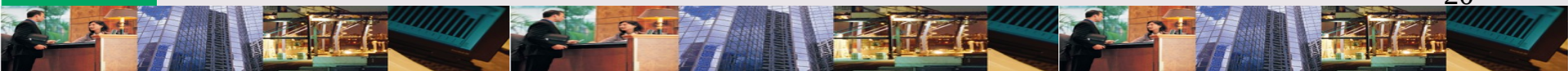
$CQ/2 = [(500/2) \times \$2] = \$ 500$

$TC = \$1,000$

REORDER POINT: Background

When using the EOQ model, identify the reorder point (ROP) reduces the probability of a stockout. To identify the reorder point, you need to know:

- ✧ Rate of usage
- ✧ Lead time required for order to be placed & received



FORMULA: Reorder Point (ROP)

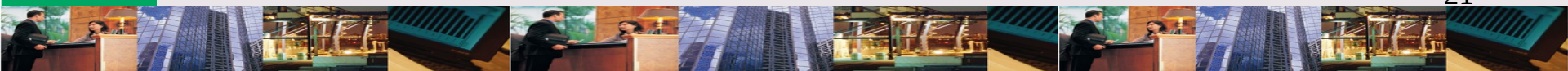
***ROP** identifies the proper time to place an order to avoid stockout.*

Reorder Point (**ROP**) equation 14.3:

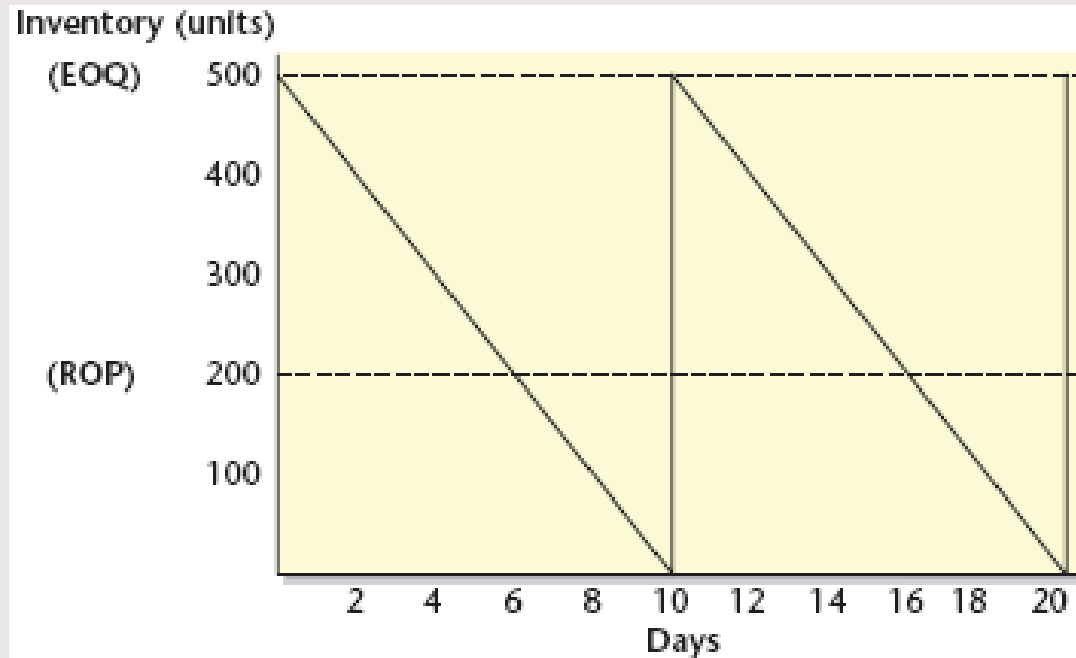
= Rate of usage x Lead time

= 50 parts per day x 4 days

= **200 parts**



REORDER POINT

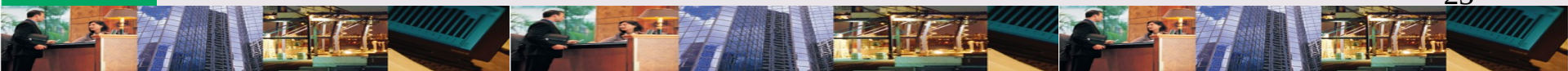


Given an optimal order quantity of 500 units, reordering should occur when 200 units remain.

EXHIBIT 14-2

SAFETY STOCK: Definition

Is extra inventory carried as insurance against fluctuations in demand.



FORMULA: Safety Stock

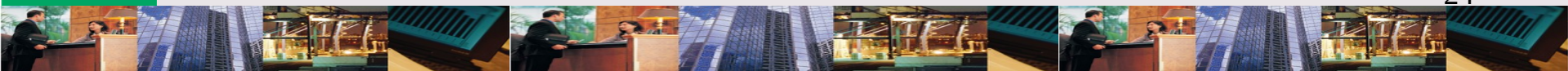
Safety stock provides a buffer to reorder point.

Safety stock:

= Lead time x (maximum – average usage)

= 4 days x (60 – 50)

= 40 parts



FORMULA: ROP + Safety Stock

Safety stock adds a buffer to reorder point.

Reorder Point (ROP) equation 14.4:

= Rate of usage x Lead time + Safety stock

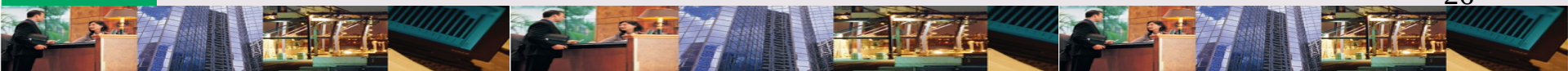
= 50 parts per day x 4 days + 40

= 240 parts

MANUFACTURING: Background

What are the EOQ and ROP for manufacturing based on information the controller provided the manager.

Ave. demand for blades	320 per day
Maximum demand for blades	340 per day
Annual demand for blades	80,000
Unit carrying cost	\$5
Setup cost	\$12,500
Lead time	20 days



EOQ & ROP: Manufacturing

$$\begin{aligned}
 EOQ &= \sqrt{\frac{2PD}{C}} \\
 &= \sqrt{\frac{2 \times 80,000 \times 12,500}{5}} \\
 &= \sqrt{400,000,000} \\
 &= 20,000 \text{ blades}
 \end{aligned}$$

Maximum usage	340
Average usage	<u>320</u>
Difference	20
Lead time	<u>×20</u>
Total safety stock	<u>400</u>

$$\begin{aligned}
 \text{Reorder point} &= (\text{Average usage} \times \text{Lead time}) + \text{Safety stock} \\
 &= (320 \times 20) + 400 \\
 &= 6,800 \text{ units}
 \end{aligned}$$

Exhibit 14-3 EOQ and Reorder Point Illustrated

EXHIBIT 14-3

The model shows that blades will be ordered in **batches of 20,000** when there are **6,800 blades remaining**.

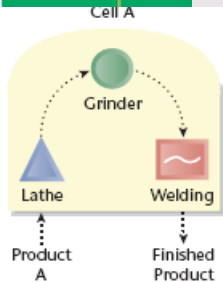
LEARNING OBJECTIVE

2

Discuss JIT
inventory
management.

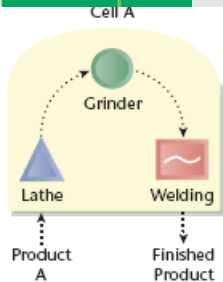
JUST-IN-TIME (JIT): Definition

Is a demand-pull manufacturing system that requires goods to be pulled through the system by present demand.



How does JIT differ from traditional inventory management?

A JIT system arranges with suppliers to deliver parts & materials just in time for production rather than on a specified predetermined schedule.



COMPARING TRADITIONAL & JIT INVENTORY MANAGEMENT

JIT

TRADITIONAL

Pull-through system

Push-through system

Insignificant inventories

Significant inventories

Small supplier base

Large supplier base

Long-term supplier contracts

Short-term supplier contracts

Cellular structure

Departmental structure

Multi-skilled labor

Specialized labor

Decentralized services

Centralized services

High employee involvement

Low employee involvement

Facilitating management style

Supervisory management style

Total quality control

Acceptable quality level

Direct tracing dominates costing

Driver tracing dominates costing

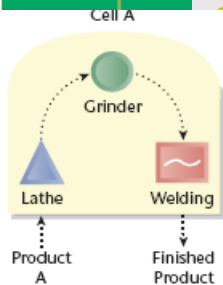
EXHIBIT 14-6

JIT: Strategic Objectives

- ✧ Increase profits
- ✧ Improve competitive position

BY

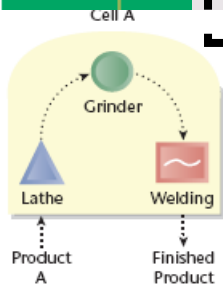
- ✧ Controlling costs
- ✧ Improving delivery performance
- ✧ Improving quality



JIT: Inventory Management Features

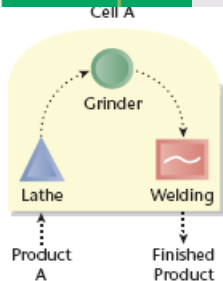
JIT manages inventory through

- ✧ Devising basic features that differ from traditional inventory systems
- ✧ Controlling setup & carrying costs
- ✧ Managing due-date performance
- ✧ Avoiding shutdown & achieving process reliability



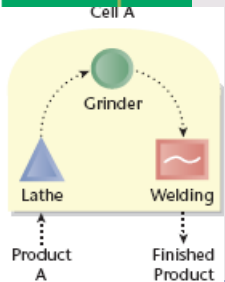
What kinds of changes
does JIT address?

Basic inventory features of JIT
address how manufacturing
facilities can be designed to
promote employee empowerment
& product quality.

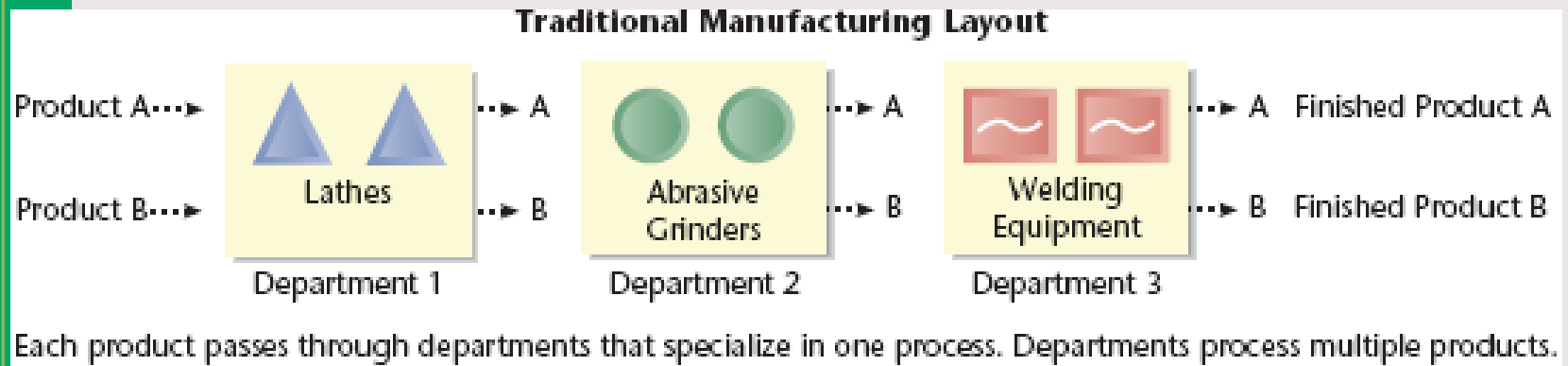


BASIC FEATURES OF JIT

- ✳ Changing plant layout to manufacturing cells
- ✳ Grouping to empower employees
- ✳ Emphasizing quality through total quality control (TQC)
- ✳ Tracing rather than allocating overhead
- ✳ Maintaining low inventory levels



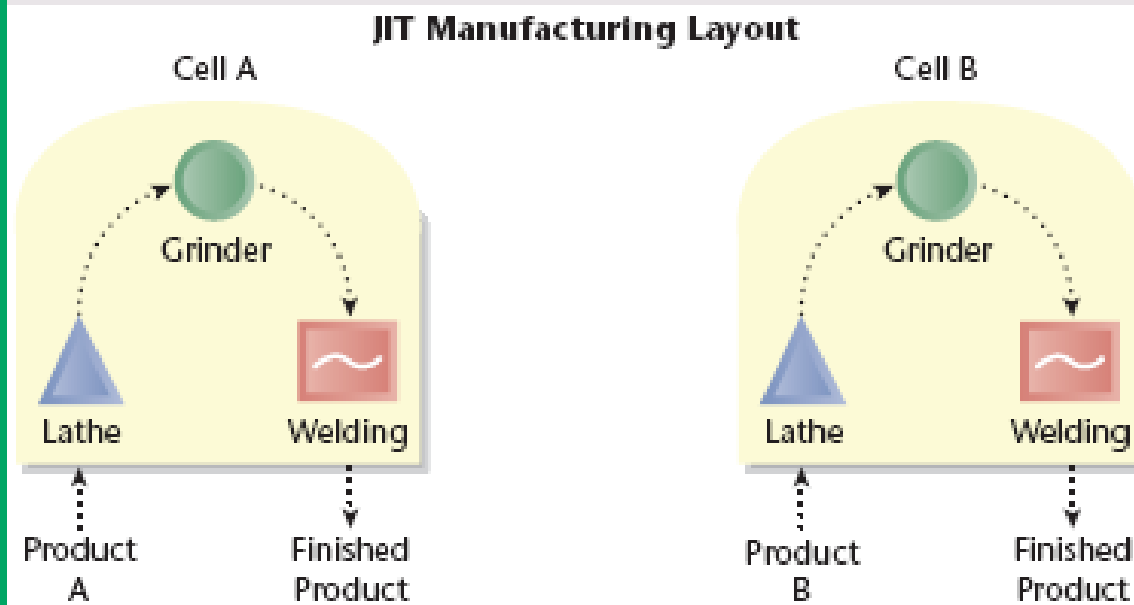
PLANT LAYOUT PATTERN: Panel A



The traditional layout pushes multiple products through departments that specialize in one activity.

EXHIBIT 14-4

PLANT LAYOUT PATTERN: Panel B



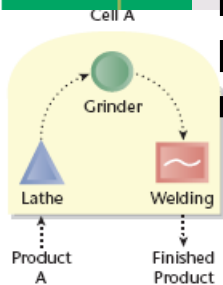
The JIT layout divides workplace into cells that complete manufacture of 1 product each.

EXHIBIT 14-4

JIT SETUP & CARRYING COSTS

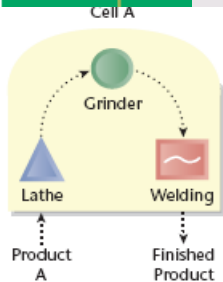
JIT uses new strategies to reduce & control setup and carrying costs of inventory

- ✳ Long-term contracts with close relationship to suppliers
- ✳ Continuous replenishment of inventory
- ✳ EDI using computers to manage inventory orders
- ✳ JIT II has supplier on-site full time



How does JIT measure supplier response?

JIT uses **due date performance** to measure a supplier's ability to respond to inventory needs.



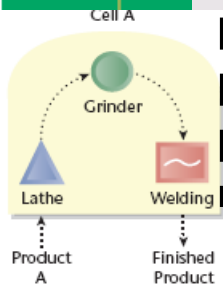
AVOIDING SHUTDOWNS: JIT

Shutdowns are caused by:

- ✳ Machine failure
- ✳ Defective material or sub-assembly
- ✳ Unavailability of material or sub-assembly

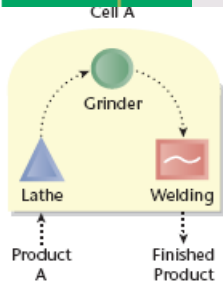
JIT response

- ✳ Total preventive maintenance
- ✳ Total quality control (TQC)
- ✳ Using the Kanban system



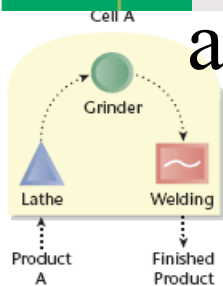
How does JIT select suppliers?

JIT selects suppliers based on performance in terms of price, quality, ability to deliver.



LIMITATIONS OF JIT

- ✳ Time is required to build sound relations with suppliers
- ✳ Workers experience stress in changing over to JIT
- ✳ Production may be interrupted because of absence of inventory supply buffer
- ✳ May place current sales at risk to achieve assurance of future sales



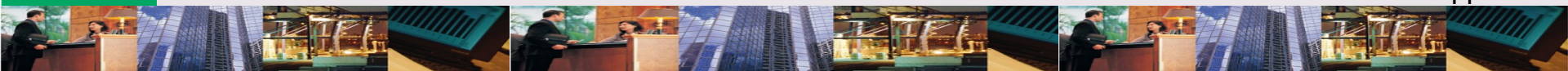
LEARNING OBJECTIVE

3

Explain the theory of constraints (TOC) & tell how it can be used to management inventory.

CONSTRAINT: Definition

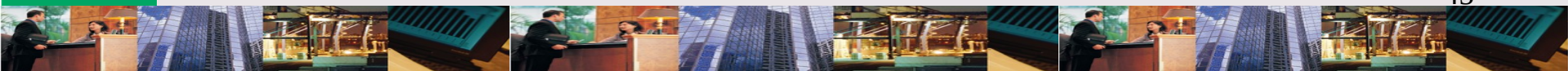
Is the limitation of
resources or product
demand.



THEORY OF CONSTRAINTS

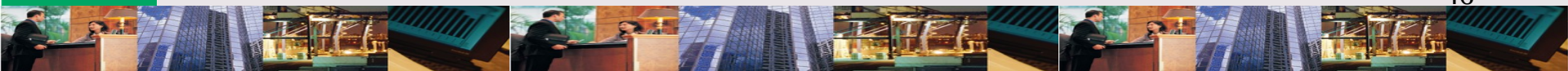
Theory of constraints (TOC) focuses on **3** measures of organizational performance:

- ✧ Throughput: rate of generating money through sales
- ✧ Inventory: money spent turning materials into throughput
- ✧ Operating expenses: money spent turning inventory into throughput



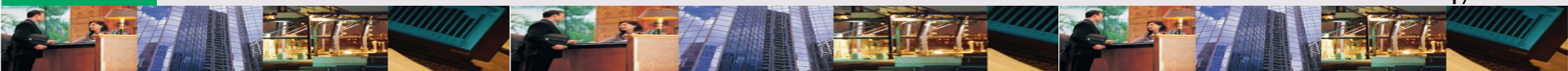
How does throughput
work?

Increasing throughput minimizes
inventory & decreases operating
expenses.



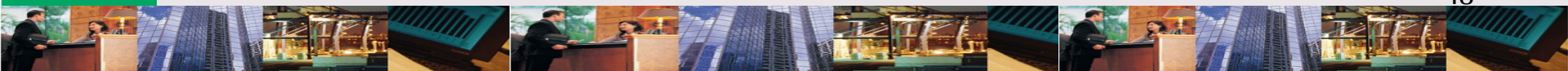
BASIC CONCEPTS: TOC

- ✧ TOC suggests that constraints (and thereby inventory) are best managed through
 - ✧ Having better, higher quality products
 - ✧ Having lower prices
 - ✧ Being responsive
 - ✧ On-time delivery
 - ✧ Shorter lead time



TOC STEPS

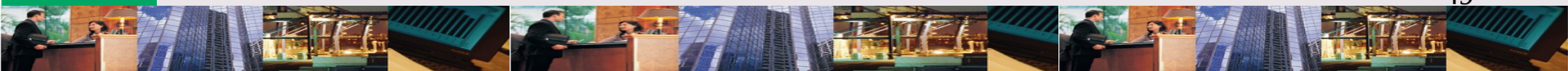
1. Identify constraints
2. Exploit binding constraints
3. Subordinate everything to decision made in #2 above
4. Elevate binding constraints
5. Repeat process



BINDING CONSTRAINTS:

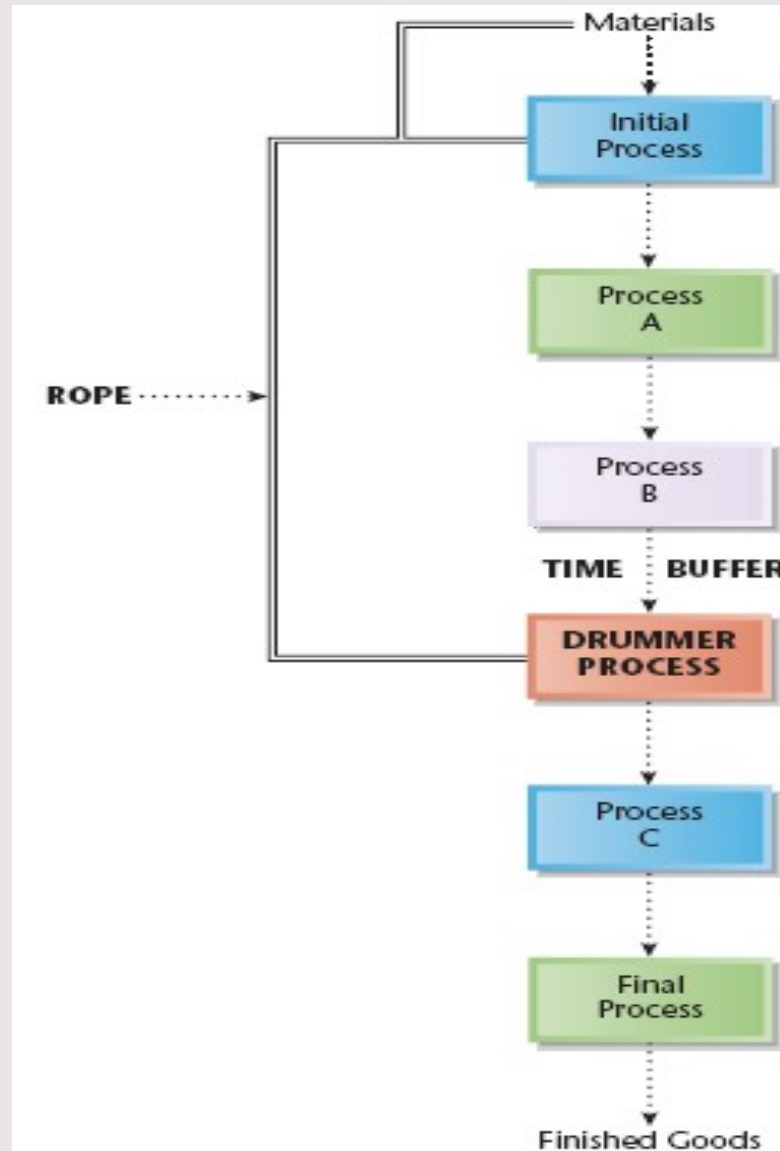
Definition

Are those constraints
whose available resources
are fully utilized.



DRUM-BUFFER-ROPE (DBR)

EXHIBIT 14-11



Additional inventory is placed before the binding constraint to give a time buffer.

CHAPTER 14

THE END